LD19694DR Chen

Title: High-order fluctuations of temperature in hot QCD matter

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Summary of revision:

We thank Referee C and Referee D for the comments, questions and suggestions. In response, we have revised the manuscript with the modification and addition highlighted (in the file "temperature-fluctuations-highlighted"). Here the revision is summarized as follows:

- We have reformulated the manuscript in alignment with the format of a regular paper in PRD, where the supplementary materials in the previous version are now included in the appendices.
- We have updated the numerical calculations in this work, added the calculations of temperature fluctuations on the phase diagram, shown in the new figures Fig. 3 and Fig. 4. The CEP, phase boundary line, and three representative freeze-out curves are shown in the phase diagram. The new figures provide more information about the relation between the temperature fluctuations and the phase structure, and clearly show how the temperature fluctuations are affected by the CEP. The relevant discussions are also added on page 4.
- We have added lattice QCD results for the pressure in Fig.8, for the comparison with our calculations.
- During the peer review process of this paper, the STAR Collaboration reported the measurements of the variance of the mean p_T fluctuations of charged particles across different collision energy [68] (arXiv:2604.06434). We have added a paragraph in the conclusion section to discuss it.
- Following the suggestion by Referee, we have added a paragraph in the conclusion section to address the relation between the temperature fluctuations and the net-proton fluctuations.
- Several discussions and modifications were made, related to the questions or comments by the referees in the following.
- Several references were added.

Answer to Referee C:

- **Comment:** *"I have read the authors' response to my previous report carefully. Unfortunately, I do not find that it addresses the two central concerns I raised, and I maintain my recommendation against publication.*

  Regarding novelty: the authors concede that the variance result appears already in Landau and Lifshitz. Their defense rests on the claim that the higher-order generalization, the timeliness relative to recent STAR data, and the negative skewness prediction collectively justify publication. These are arguments about interest and context, not about mathematical substance. The *higher-order cumulants follow by routine differentiation once the state function W is defined, and no argument is made that this step is technically nontrivial.*

Regarding experimental connection: the authors' response to my request for at least a semi-realistic estimate of competing contributions is, in essence, that doing so would be too large a project for a single paper. I accept that a full hydrodynamic study is beyond the scope of a Letter. However, the paper as written makes the strong claim that the state function W 'directly connects to the mean transverse momentum fluctuations measured in heavy-ion collisions.' This claim is not qualified in the abstract or introduction, and no order-of-magnitude estimate is offered to support the assertion that the thermodynamic signal is accessible above the background from resonance decays and radial flow.

In summary, the response is primarily argumentative. No new calculations have been performed, no claims have been revised, and no major changes to the manuscript are documented. I cannot recommend publication in Physical Review D. "

Answer: Thanks for the comments. We have made a major revision. We have performed new calculations and made many changes to the manuscript. Summary of the revision is presented in the beginning of this response. In particular, we have added a caveat about the limitation in this work when it is connected to the experimental data at the end of Section IV. "There is a caveat when the results in Fig. 5 are compared with experimental data. As we have discussed above, the results in Fig. 5 are obtained by evaluating the temperature fluctuations in thermal equilibrium at the chemical freeze-out. To facilitate a better comparison with experiments, it is desirable to include other effects, such as the initial state geometry fluctuations, flow fluctuations, etc., in a dynamical evolution. This is beyond the scope of this paper, which will be investigated in the future."

With a such major revision, we hope you could reconsider our manuscript. Thanks.

Answer to Referee D:

- **Comment:** *"In the submitted paper, the authors attempt to establish temperature fluctuations as a new, experimentally accessible probe of QCD thermodynamics and the QCD phase structure. They develop a theoretical framework to compute temperature fluctuations of arbitrary order in hot QCD matter and connect these to event-by-event fluctuations of mean transverse momentum $\langle p_t \rangle$. The predictions provided have relevance for the RHIC beam energy scan program as well as physics at FAIR or NICA.*

  Conceptually new is the introduction of the thermodynamic state function W which is motivated as the natural thermodynamic potential for heavy-ion collision observables. Temperature cumulants are then consistently derived using a low-energy effective theory with FRG methods. These are connected to observables via the conjectured proportionality between mean p_t and T.

  The authors find a strong suppression of temperature fluctuations in the QGP which is assigned to a large heat capacity. Throughout the phase diagram, a negative c_3 is assigned to an asymmetric temperature distribution with small fluctuations in the high-T region and a broadening for low T. In analogy to net-proton fluctuations, strong sensitivity is found near the crossover, including sign change for higher order cumulants.

In my opinion, the strength of this work lies in the introduction of $W(S, V, \mu)$ as the relevant thermodynamic potential and the attempt to bridge cleanly controllable thermodynamics with experimental observables. The obtained results are methodologically sound and seemingly relevant for experiment. The phenomenological realization, however, is nontrivial and must be made clear before I can recommend the manuscript for publication. "

A: Thank you for the comments.

- **Comment:** *"The main limitation I see is that temperature fluctuations (unlike net-proton cumulants) are not a precise observable but rather have to rely on $\langle p_t \rangle = aT$, a model-dependent relation. Competition with other established probes like the net-proton fluctuations, or e.g. dileptons, is obvious and the authors must justify why they believe that temperature fluctuations add unique (and new?) information. "*

  A: The observable of temperature fluctuations is complementary to those of net-proton fluctuations or the dileptons. Since it is far more difficult to use the measurement of dileptons to detect the criticality or the CEP due to its constraint of statistics in experiments, here we compare the temperature fluctuations and net-proton fluctuations. Firstly, on the QCD phase diagram in terms of the temperature and baryon chemical potential, the net-proton fluctuations are related to the direction of the baryon chemical potential and the temperature fluctuations are related to the direction of temperature. A comprehensive study of the QCD criticality in the two directions is helpful to investigate the two-dimensional structure of QCD phase diagram. Secondly, using the net-proton fluctuations to search for the CEP relies on the coupling between the conserved charges, e.g., the baryon number, and the critical mode, i.e., the sigma mode of CEP, while the temperature fluctuation does not rely on such coupling, and it represents directly the properties of critical phenomena. Finally, partly due to the reason mentioned in the second point, the critical exponents of temperature fluctuations are the well-know universal exponents. For example the variance of the temperature fluctuations is just the inverse of the heat capacity as shown in Eq.(11). However, the critical exponents of baryon or proton fluctuations are not well determined, and assumptions or models are required in the computation of these exponents.

  Furthermore, if some signal of CEP in the net-proton fluctuations is observed in experiments in the future, another independent signal from other observables, e.g., the temperature fluctuations will certainly be welcome and could potentially consolidate the finding. In the revised version, we have added a paragraph in the conclusion section to address the relation between the temperature fluctuations and the net-proton fluctuations, beginning with "Furthermore, we also would like to comment on...".

- *"Besides this main issue, I have the following concerns:*

  Question: 1. *Cumulants are calculated in thermodynamic equilibrium, no dynamical modeling is done which further limits the applicability to data. Figs. 1-3 show calculations at constant chemical potential or along parametrized freeze-out curves. This should at least be clearly stated and its limitations must be emphasized."*

  A: We agree with the comment. We have added a paragraph on page 4 to address this issue, beginning with "There is a caveat when the results in...".

- **2.** *"According to ref. [37], the value of "a" (the proportionality constant between p_t and T-effective) varies by about 4-6%, depending on details of the model and on the freeze-out temperature. I would like to know from the authors about the impact of the hydrodynamic expansion, flow fluctuations, non-thermal fluctuations, or event-by-event changes of the system size on the value of "a" – in general, a stronger justification between the relation between mean p_t and T fluctuations in realistic heavy-ion collisions. I am not convinced that these fluctuations would cancel by considering the cumulant ratios and their impact must be assessed carefully. "*

  A: As the hydrodynamic simulations show in Ref. [38] ([37] in the previous version, arXiv: 1908.09728), while the $\langle p_T \rangle$ and the effective T are sensitive to details of the hydrodynamic modeling, e.g., the initial density profile, transport coefficients, equation of state, etc., the model dependence disappears when considering the ratio $\langle p_T \rangle / T$, i.e., the proportionality constant a. Moreover, it is also found that a is a constant across different centrality and different collision energy from RHIC to LHC. The error on the a is about 4%, arising from the uncertainty in the freeze-out temperature. Note that the error 6% from a stiff equation of state was not taken into account, since it overestimates $\langle p_t \rangle$ by 40% in comparison to the experimental LHC data.

  Given the insensitivity of the ratio a to the hydrodynamic modeling mentioned above, it is reasonable to assume that the almost constant a also holds in event-by-event collisions, since every event is independent of each other. However, this assumption should be verified in the simulations of dynamical models in the near future. In the revised version, We have added a paragraph at the end of Sec. III to discuss it, beginning with "Note that results of hydrodynamic simulations in...".

- **3.** *"The statement and basis for the use of W, that entropy S scales directly with the charged particle multiplicity (page 2) is only approximately true. Fluctuations from decays, acceptance, etc. play a role that should be emphasized. "*

  A: Thanks for the comment. We have added a sentence at the end of Sec. II to emphasize it.

- **4.** *"The authors motivate their study partly by the sensitivity of their fluctuations to criticality in the QCD phase diagram. This connection should be made more explicit, e.g., by investigating the cumulants near a critical point. "*

  A: Thanks for the suggestion. In the revised version, we have added the calculations of temperature fluctuations on the phase diagram, shown in the new figures Fig. 3 and Fig. 4. The CEP, phase boundary line, and three representative freeze-out curves are shown in the phase diagram. The new figures provide more information about the relation between the temperature fluctuations and the phase structure, and clearly show how the temperature fluctuations are affected by the CEP. The relevant discussions are also added on page 4.

- **5.** *"The LEFT model used in this study requires stronger (or any) motivation. I would urge the authors to describe properties of the model, how does it relate to lattice data, does it have a critical point and if yes, where? Is the step-like behavior in Fig.2 at the highest value of*

μ related to criticality? This is an essential part of this work and should not be completely outsourced to supplemental material (which also does not inform much about this) or previous publications. "

A: We have added lattice QCD results for the pressure in Fig.8, which are compared with the results of LEFT. It is found that the LEFT is consistent with the lattice when $T \lesssim 230$ MeV, which is the temperature range most concerned in this work, while the LEFT overshoots a bit the lattice results in the region of high temperature. There is a critical end point in the LEFT, whose location is given in Eq.(12), shown also in the phase diagram in Fig. 3 and Fig. 4. After a thorough analysis of the temperature fluctuations in the phase diagram in Fig. 3 and Fig. 4, one finds that the chiral crossover becomes sharper and sharper with the increase of μ_B, and there are some nontrivial structure developed near the phase boundary when the system is approaching toward the CEP. The step-like behavior in Fig.2 stems from the nontrivial structure and the increasingly sharp crossover with the increase of μ_B. In the revised version, we have added relevant discussions.

- **Comment:** *"Overall, while I definitely see value in this study, I think these issues require some significant modification before I can recommend the paper for publication in Physical Review D."*

  A: Thanks for the comment. We have done further calculations and revised the manuscript accordingly to address all these issues carefully.

- **Comment:** *"Justification for/against Letter:*

  The manuscript presents an interesting and conceptually new framework for describing temperature fluctuations in hot QCD matter and establishes a connection to mean transverse momentum fluctuations. However, the results are mainly theoretical and rely on assumptions whose validity in realistic heavy-ion collisions (particularly the mapping between event-by-event mean transverse momentum fluctuations and temperature fluctuations) has not been demonstrated quantitatively. The lack of validation within a dynamical framework or direct comparison to experimental observables limits the immediate impact and applicability of the results. While the work is suitable for publication, it does not meet the level of broad, immediate significance typically required for a PRD Letter and would be more appropriate as a Regular PRD Article."

  A: We respect this justification. The revised version is a format of regular PRD article.